

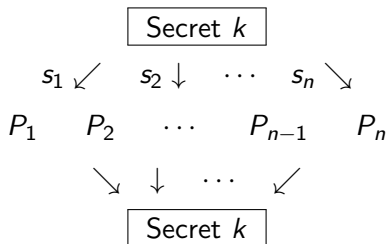
Secret Sharing

Lecture 19 — Access Structures

SEC 595 — Encrypted Computing

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Splitting a secret



Down: a dealer hands share s_i to participant P_i . **Up:** a subset pools its shares and reconstructs.

The reconstructing group need not be everybody. So **who**? Which subsets may recover k , and which must learn nothing?

Security here is **unconditional** — no hardness assumption.

Fix the **participants** $\{P_1, P_2, \dots, P_n\}$.

A **collection** is a set of subsets of participants:

$$\mathcal{A} \subseteq 2^{\{P_1, \dots, P_n\}}.$$

Keep the levels apart: P_1 is a participant, $\{P_1\}$ is a set of participants and may belong to $\mathcal{A}$.

Definition

$\mathcal{A}$ is **monotone** if

$$B \in \mathcal{A} \text{ and } B \subseteq C \implies C \in \mathcal{A}.$$

Monotonicity is forced, not assumed: if B can recover the secret, then so can B plus a newcomer — it simply ignores the newcomer.

Definition

$B \in \mathcal{A}$ is **minimal** if no proper subset of B lies in $\mathcal{A}$. The minimal sets form $\mathcal{A}^-$.

Proposition

A monotone collection is determined by its minimal sets:

$$\mathcal{A} = \{ C : B \subseteq C \text{ for some } B \in \mathcal{A}^- \}.$$

Proof: every $C \in \mathcal{A}$ contains a smallest subset lying in $\mathcal{A}$, which is minimal.

So an access structure is specified by a short list.

Definition

An **access structure** is a monotone collection $\mathcal{A}$ of *non-empty* subsets of $\{P_1, \dots, P_n\}$.

Sets in $\mathcal{A}$ are **authorized**; sets outside $\mathcal{A}$ are **unauthorized**.

$\emptyset \notin \mathcal{A}$: nobody, holding nothing, recovers the secret.

Every subset is one or the other, and the scheme owes something to each:

- authorized sets **must** recover the secret;
- unauthorized sets **must** learn nothing.

Example — monotone closure

$$n = 3, \quad \mathcal{A} = \{ \{P_1, P_2\}, \{P_1, P_3\} \}.$$

- a. Is $\mathcal{A}$ monotone?
- b. Find the smallest monotone collection containing $\mathcal{A}$. How many sets does it have?
- c. Which of its sets are minimal?

(t, n) -threshold scheme

Any t of the n participants can recover the secret, and no fewer:

$$\mathcal{A}_{t,n} = \{ B \subseteq \{P_1, \dots, P_n\} : |B| \geq t \}, \quad 1 \leq t \leq n.$$

Monotone, since $|C| \geq |B|$ when $B \subseteq C$; and $t \geq 1$ excludes $\emptyset$.

Minimal sets: the $\binom{n}{t}$ sets of size t . $|\mathcal{A}_{t,n}| = \sum_{i=t}^n \binom{n}{i}$.

(n, n) : unanimity. $(1, n)$: everyone alone — replication, no confidentiality.

Not every policy is a count. Participants A, B, C, D :

only A and B together can recover.

$\mathcal{A}^- = \{\{A, B\}\}$, so $\mathcal{A}$ is the four supersets of $\{A, B\}$.

No threshold expresses this: $\{A, B\}$ and $\{C, D\}$ have the same size, but only one is authorized.

Example — counting authorized sets

- a. Describe (n, n) , $(1, n)$ and $(2, 4)$ in words. How many authorized sets does each have?
- b. Count the authorized sets of $(3, 5)$ and of $(2, 5)$.
- c. Participants A, B, C, D ; $\mathcal{A}$ is every set containing $\{A, B\}$. List $\mathcal{A}$ and the unauthorized sets. Why is $\mathcal{A}$ not $\mathcal{A}_{t,4}$ for any t ?

Definition

A **distribution scheme** is a pair $\Sigma = \langle \Pi, \mu \rangle$:

$$\Pi : K \times R \longrightarrow K_1 \times K_2 \times \cdots \times K_n,$$

with μ a probability distribution on the random strings R .

- K — the public **domain of the secret**; $k \in K$.
- $\Pi(k, r) = (s_1, \dots, s_n)$; share $s_i \in K_i$ goes to P_i only.
- $\Pi(k, r)_B$ — the shares held by the members of B .

The dealer then discards r .

Randomness is not optional

With no randomness each share is a fixed function of k , and an unauthorized set's shares must be *constant* in k .

Proposition

If $|R| = 1$ and $\langle \Pi, \mu \rangle$ implements $\mathcal{A}$, then every minimal set of $\mathcal{A}$ is a singleton.

Proof: for minimal $B \ni P_i, P_j$, the shares of $B \setminus \{P_i\}$ and $B \setminus \{P_j\}$ are constant, so all of B 's are, and RECON_B cannot return k .

So no threshold scheme with $t \geq 2$ is deterministic.

Let K be finite with $|K| \geq 2$. $\langle \Pi, \mu \rangle$ is a **secret sharing scheme** for $\mathcal{A}$ if it is correct and private.

Correctness

For every $B = \{P_{i_1}, \dots, P_{i_m}\} \in \mathcal{A}$ there is a **reconstruction function** $\text{RECON}_B : K_{i_1} \times \dots \times K_{i_m} \rightarrow K$ with

$$\Pr_{r \sim \mu} [\text{RECON}_B(\Pi(k, r)_B) = k] = 1 \quad \text{for all } k \in K.$$

Probability exactly 1: no noise, no failure.

RECON_B sees only B 's shares, and there is one per authorized set.

Privacy

For every $T \notin \mathcal{A}$, all secrets $a, b \in K$, and every share vector $\langle s_j \rangle_{P_j \in T}$,

$$\Pr_{r \sim \mu} [\Pi(a, r)_T = \langle s_j \rangle] = \Pr_{r \sim \mu} [\Pi(b, r)_T = \langle s_j \rangle].$$

The shares seen by an unauthorized set have the **same distribution** for every secret: the view of T is independent of k .

Equivalently, for any prior on k : $\Pr[k \mid \Pi(k, r)_T = \sigma_T] = \Pr[k]$.

The privacy condition names no adversary, no running time and no hardness assumption. It is **information-theoretic** (perfect) security.

Contrast L6–L15: Regev and GSW rest on LWE, ElGamal on DDH, Paillier on composite residuosity. Here the advantage is not negligible — it is zero.

- No parameter grows with λ ; better algorithms or quantum computers change nothing.
- The price is size: a non-redundant participant's share domain satisfies $|K_i| \geq |K|$.
- The guarantee covers the shares only — not a cheating dealer or a forged share.

$K = R = \mathbb{Z}_2$, $n = 2$, r uniform: $\Pi(k, r) = (r, k \oplus r)$.

	$r = 0$	$r = 1$
$k = 0$	$(0, 0)$	$(1, 1)$
$k = 1$	$(0, 1)$	$(1, 0)$

Correct: $\text{RECON}_{\{P_1, P_2\}}(s_1, s_2) = s_1 \oplus s_2$ in every cell.

Private: in each row, s_1 and s_2 each take 0 and 1 once — a single share is a uniform bit whatever k is.

It implements $\mathcal{A}_{2,2} = \{\{P_1, P_2\}\}$, with $|K_i| = |K|$.

Scheme

To share $k \in \mathbb{Z}_p$ among n parties: draw $s_1, \dots, s_{n-1}$ uniformly from $\mathbb{Z}_p$ and set

$$s_n = k - (s_1 + \dots + s_{n-1}) \bmod p.$$

$$\text{RECON}(s_1, \dots, s_n) = s_1 + \dots + s_n \bmod p.$$

Any $n - 1$ shares are uniform and independent of k ; all n sum to k .

It implements $\mathcal{A}_{n,n}$. For $p = 2$, $n = 2$ it is the XOR scheme.

Example — the XOR scheme

$\Pi(k, r) = (r, k \oplus r)$ over $\mathbb{Z}_2$.

- a. Verify correctness and privacy cell by cell.
- b. Change μ to $\Pr[r = 0] = \frac{3}{4}$. Name a participant and a share value whose probability differs between $k = 0$ and $k = 1$.
- c. Why does the other participant notice nothing?

Example — additive sharing

$K = \mathbb{Z}_5$, $n = 3$: draw s_1, s_2 uniformly, $s_3 = k - s_1 - s_2 \bmod 5$.

- a. Write RECON for the full set.
- b. Show that any two participants hold any given pair of values with probability $\frac{1}{25}$, whatever k is.
- c. Which access structure is implemented if the dealer draws only s_1 and sets $s_2 = 0$?

Variation	What it changes
Threshold	$\mathcal{A}$ depends only on $ B $
General access	$\mathcal{A}$ given by its minimal sets
Multi-span	several schemes share one s_i
Multisecret	one scheme, several secrets
Verifiable	dealer and participants may cheat
Dynamic	participants added, updated, removed
Quantum	shares are quantum states

Each is still judged by correctness and privacy. Constructions for threshold and general structures: Shamir, CRT, Brickell (Lecture 20).

- 1 An access structure $\mathcal{A}$ is a monotone collection of non-empty sets of participants, determined by its minimal sets. Inside $\mathcal{A}$: authorized; outside: unauthorized.
- 2 (t, n) -threshold: $\mathcal{A}_{t,n} = \{B : |B| \geq t\}$, with $\sum_{i \geq t} \binom{n}{i}$ authorized sets.
- 3 A distribution scheme $\langle \Pi, \mu \rangle$ maps $K \times R \rightarrow K_1 \times \dots \times K_n$; the randomness is essential.
- 4 Correctness: each authorized B has RECON_B , exact with probability 1. Privacy: each unauthorized T sees the same share distribution for every secret.
- 5 Privacy is information-theoretic; the price is shares at least as large as the secret. XOR and additive sharing implement $\mathcal{A}_{n,n}$.